

Rational Points on Elliptic Curves
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1 Geometry and Arithmetic

Problem 1

1. If P and Q are distinct rational points in the (x, y) plane, prove that the line connecting them is a rational line.
2. If L_1 and L_2 are distinct rational lines in (x, y) plane, prove that their intersection is a rational point (or empty).

Proof. Suppose P and Q are distinct rational points in the (x, y) plane. Let $P = (p_1, p_2)$ and $Q = (q_1, q_2)$. We seek an equation of a line with rational coefficients such that P and Q are solutions. There are two cases to consider. First, suppose $q_1 = p_1$. Since $P \neq Q$ we must have $p_2 \neq q_2$. Then the vertical line $x = p_1$ has rational coefficients and is satisfied by both P and Q , since $p_1 = q_1$ is rational. Now, suppose $q_1 \neq p_1$. Set $m = \frac{q_2 - p_2}{q_1 - p_1}$ and $b = p_2 - mp_1$. Both are rational since p_1, p_2, q_1, q_2 are rational and $q_1 - p_1 \neq 0$. Substituting the coordinates of P into $y = mx + b$ gives

$$p_2 = \left(\frac{q_2 - p_2}{q_1 - p_1} \right) p_1 + b,$$

which holds by our choice of b . Substituting Q gives

$$mq_1 + b = mq_1 + p_2 - mp_1 = m(q_1 - p_1) + p_2 = (q_2 - p_2) + p_2 = q_2,$$

so Q is also a solution. ■

Proof. Suppose L_1 and L_2 are distinct rational lines in the (x, y) plane. Write L_1 and L_2 as

$$y = m_1x + b_1 \text{ and } y = m_2x + b_2 \text{ respectively,}$$

where m_1, m_2, b_1, b_2 are rational. A point (x, y) lies on both lines exactly when it solves the system

$$\begin{bmatrix} -m_1 & 1 \\ -m_2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Let $A = \begin{bmatrix} -m_1 & 1 \\ -m_2 & 1 \end{bmatrix}$, so $\det A = -m_1 + m_2 = m_2 - m_1$. Suppose $m_1 \neq m_2$, so $\det A \neq 0$ and A is invertible. By Cramer's rule the system has the unique solution

$$x = \frac{1}{\det A} \begin{vmatrix} b_1 & 1 \\ b_2 & 1 \end{vmatrix} = \frac{b_1 - b_2}{m_2 - m_1}, \quad y = \frac{1}{\det A} \begin{vmatrix} -m_1 & b_1 \\ -m_2 & b_2 \end{vmatrix} = \frac{m_2 b_1 - m_1 b_2}{m_2 - m_1}.$$

Since m_1, m_2, b_1, b_2 are rational and $m_2 - m_1 \neq 0$, both x and y are rational, so the intersection point is rational. ■

Problem 2

Let C be the conic section given by the equation

$$F(x, y) = ax^2 + bxy + cy^2 + dx + ey + f = 0,$$

and let δ be the determinant

$$\det \begin{bmatrix} 2a & b & d \\ b & 2c & e \\ d & e & 2f \end{bmatrix}.$$

1. Show that if $\delta \neq 0$, then C has no singular points. That is shows that there are no points (x, y) where

$$F(x, y) = \frac{\partial F}{\partial x}(x, y) = \frac{\partial F}{\partial y} = 0,$$

2. Conversely, show that if $\delta = 0$ and $b^2 - 4ac \neq 0$, then there is a unique singular point on C .
3. Let L be the line $y = \alpha x + \beta$ with $\alpha \neq 0$. Show that the intersection of L and C consists of either zero, one, or two points.
4. Determine the conditions on the coefficients to ensure that the intersection $L \cap C$ consists of exactly one point. What is the geometric significance of these conditions? (Note. That there will be more than one case.)

Proof. Let (x, y) be a singular point of C , so that $F = 0$, $\frac{\partial F}{\partial x} = 0$, and $\frac{\partial F}{\partial y} = 0$. Since

$$\frac{\partial F}{\partial x} = 2ax + by + d \text{ and } \frac{\partial F}{\partial y} = bx + 2cy + e,$$

and since

$$2F = \begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} 2a & b & d \\ b & 2c & e \\ d & e & 2f \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix},$$

the three conditions $\frac{\partial F}{\partial x} = \frac{\partial F}{\partial y} = 0$ and $dx + ey + 2f = 0$ give

$$\begin{bmatrix} 2a & b & d \\ b & 2c & e \\ d & e & 2f \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The vector $(x, y, 1) \neq \theta$ so $\delta = 0$ which is a contradiction. ■

Proof. Suppose $\delta = 0$ and $b^2 - 4ac \neq 0$. As above, a point (x, y) is singular iff

$$\begin{bmatrix} 2a & b & d \\ b & 2c & e \\ d & e & 2f \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first two rows say

$$\begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -d \\ -e \end{bmatrix},$$

and the determinant of this 2×2 system is $4ac - b^2 \neq 0$, so it has a unique solution (x, y) . Since $\delta = 0$ but the first two rows are independent, the third row is a combination of them, so it vanishes on $(x, y, 1)$. Thus (x, y) satisfies $dx + ey + 2f = 0$, and is the unique singular point on C . ■

Proof. We can write $y = \alpha x + \beta$ since L is given. Then substituting into C we find

$$F(x, \alpha x + \beta) = ax^2 + bx(\alpha x + \beta) + c(\alpha x + \beta)^2 + dx + e(\alpha x + \beta) + f = 0.$$

Collecting gives a quadratic in x ,

$$Ax^2 + Bx + C = 0,$$

where

$$A = a + b\alpha + c\alpha^2, \quad B = b\beta + 2c\alpha\beta + d + e\alpha, \quad C = c\beta^2 + e\beta + f.$$

This clearly has 0, 1, or 2 roots. ■

Proof. With A, B, C as above, $L \cap C$ is one point exactly when $Ax^2 + Bx + C = 0$ has one solution. There are two cases. If $A \neq 0$, the quadratic has one root iff

$$B^2 - 4AC = 0.$$

If $A = 0$, that is $a + b\alpha + c\alpha^2 = 0$, the equation is linear and has one root iff $B \neq 0$. ■

Problem 3

Let C be the conic given by the equation

$$F(x, y) = x^2 - 3xy + 2y^2 - x + 1 = 0.$$

1. Check that C is non-singular (cf. the previous exercise).
2. Let L be the line $y = \alpha x + \beta$. Suppose that the intersection $L \cap C$ contains the point $P_0 = (x_0, y_0)$. Assuming that $L \cap C$ consists of two distinct points, find the second point of $L \cap C$ in terms of α, β, x_0, y_0 .
3. If L is a rational line and P_0 is a rational point (i.e., $\alpha, \beta, x_0, y_0 \in \mathbb{Q}$), prove that the second point of $L \cap C$ is also a rational point.

Proof. We have

$$A = \begin{bmatrix} 1 & -\frac{3}{2} & -\frac{1}{2} \\ -\frac{3}{2} & 2 & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix},$$

and note $\det(A) = -\frac{3}{4} \neq 0$, thus C is non-singular. ■

Proof. We can write $y = \alpha x + \beta$ since L is given. With $a = 1, b = -3, c = 2, d = -1, e = 0, f = 1$, substituting into C gives

$$F(x, \alpha x + \beta) = ax^2 + bx(\alpha x + \beta) + c(\alpha x + \beta)^2 + dx + e(\alpha x + \beta) + f = 0.$$

Collecting gives a quadratic in x ,

$$Ax^2 + Bx + C = 0,$$

where

$$A = a + b\alpha + c\alpha^2, \quad B = b\beta + 2c\alpha\beta + d + e\alpha, \quad C = c\beta^2 + e\beta + f.$$

We know $A \neq 0$ since $L \cap C$ consists of two points, so the quadratic has exactly two roots x_0, x_1 and factors as

$$Ax^2 + Bx + C = A(x - x_0)(x - x_1).$$

Expanding the right side gives $Ax^2 - A(x_0 + x_1)x + Ax_0x_1$. Comparing the coefficient of x with B gives $-A(x_0 + x_1) = B$, thus

$$x_1 = -\frac{B}{A} - x_0,$$

and the y -coordinate is $y_1 = \alpha x_1 + \beta$. Thus the second point is

$$P_1 = \left(-\frac{B}{A} - x_0, \alpha \left(-\frac{B}{A} - x_0 \right) + \beta \right).$$

This is clearly rational answering part (iii). ■

Problem 3

Let C be the conic given by the equation

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1. Check that C is non-singular (cf. the previous exercise).
2. Let L be the line $y = \alpha x + \beta$. Suppose that the intersection $L \cap C$ contains the point $P_0 = (x_0, y_0)$. Assuming that $L \cap C$ consists of two distinct points, find the second point of $L \cap C$ in terms of α, β, x_0, y_0 .
3. If L is a rational line and P_0 is a rational point (i.e., $\alpha, \beta, x_0, y_0 \in \mathbb{Q}$), prove that the second point of $L \cap C$ is also a rational point.

Problem 4

Find all primitive integral right triangles whose hypotenuse has length less than 30.

Proof. We start by finding all relatively prime integers > 0 whose squares sum to ≤ 30 with $m < n$ and m and n are not both odd.

1. $1^2 + 2^2 = 5 \leq 30$
2. $1^2 + 4^2 = 17 \leq 30$
3. $2^2 + 3^2 = 13 \leq 30$
4. $2^2 + 5^2 = 29 \leq 30$
5. $3^2 + 4^2 = 25 \leq 30$

For each pair (m, n) , we get a triple $X = n^2 - m^2$, $Y = 2mn$, $Z = n^2 + m^2$ so

1. $(m, n) = (1, 2)$ we have $(3, 4, 5)$
2. $(m, n) = (1, 4)$ we have $(15, 8, 17)$
3. $(m, n) = (2, 3)$ we have $(5, 12, 13)$
4. $(m, n) = (2, 5)$ we have $(21, 20, 29)$
5. $(m, n) = (3, 4)$ we have $(7, 24, 25)$

Problem 5

Describe all rational points on the circle $x^2 + y^2 = 2$.

Proof. Let $F(x, y) = x^2 + y^2 - 2 = 0$. Consider the line $L = y = -x$. Now we compute the line through $(1, 1)$ and $y = -x$. Now parametrizing $y = -x$ gives the set of points of the form $(t, -t)$ where $t \in \mathbb{Q}$. So

$$y = \frac{-t-1}{t-1}(x-1) + 1.$$

Notice, since we take $x \neq 1$ the line is well defined. Then $F\left(x, \frac{-t-1}{t-1}(x-1) + 1\right)$ we have

$$\frac{2(x-1)((t^2+1)x-2t)}{(t-1)^2} = 0.$$

Then, $x = 1$ corresponds to the known point $(1, 1)$ so we have

$$(t^2 + 1)x - 2t = 0,$$

so $x = \frac{2t}{t^2+1}$ and $y = \frac{t^2-1}{t^2+1}$. So the set of all rational points on the circle $x^2 + y^2 = 2$ is

$$\left\{ \left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1} \right) : t \in \mathbb{Q} \right\} \cup \{(1, 1)\}.$$

Problem 6

1. Let a, b, c, d, e, f be non-zero real numbers. Use the substitution $t = \tan(\theta/2)$ to transform the integral

$$\int \frac{a + b \cos \theta + c \sin \theta}{d + e \cos \theta + f \sin \theta} d\theta$$

into the integral of a rational function of t .

2. Evaluate the integral

$$\int \frac{a + b \cos \theta + c \sin \theta}{1 + \cos \theta + \sin \theta} d\theta.$$

Proof. Substituting $\cos \theta = \frac{1-t^2}{1+t^2}$, $\sin \theta = \frac{2t}{1+t^2}$, and $d\theta = \frac{2}{1+t^2} dt$ we have

$$\frac{a + b \cos \theta + c \sin \theta}{d + e \cos \theta + f \sin \theta} d\theta = \frac{a + b \frac{1-t^2}{1+t^2} + c \frac{2t}{1+t^2}}{d + e \frac{1-t^2}{1+t^2} + f \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt.$$

Multiplying numerator and denominator of the fraction by $1 + t^2$ gives

$$= \frac{a(1+t^2) + b(1-t^2) + 2ct}{d(1+t^2) + e(1-t^2) + 2ft} \cdot \frac{2}{1+t^2} dt = \frac{2[(a-b)t^2 + 2ct + (a+b)]}{(1+t^2)[(d-e)t^2 + 2ft + (d+e)]} dt.$$

Proof. Setting $d = e = f = 1$ so

$$\begin{aligned} \int \frac{2[(a-b)t^2 + 2ct + (a+b)]}{(1+t^2)[2(t+1)]} dt &= \int \frac{(a-b)t^2 + 2ct + (a+b)}{(t+1)(t^2+1)} dt \\ &= \frac{a+c}{2} \ln |t+1| + \frac{a-b}{4} \ln(t^2+1) + (b+c) \arctan t + K. \end{aligned}$$

Problem 7

For each of the following conics, either find a rational point or prove that there are no rational points.

1. $x^2 + y^2 = 6$
2. $3x^2 + 5y^2 = 4$
3. $3x^2 + 6y^2 = 4$.

Proof. Suppose there exists $\frac{a}{b}, \frac{x}{y} \in \mathbb{Q}$ such that $\left(\frac{a}{b}\right)^2 + \left(\frac{x}{y}\right)^2 = 6$. Then

$$\left(\frac{a}{b}\right)^2 + \left(\frac{x}{y}\right)^2 = 6 \iff a^2 y^2 + x^2 b^2 = b^2 y^2 6.$$